8R12-3

Official Review of Submission10802 by Reviewer 8R12

Official Reviewby Reviewer 8R1205 Jul 2026, 05:14 (modified: 08 Jul 2026, 15:46)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 8R12, AuthorsRevisions

Paper Summary:

This paper targets the view (attributed to prior work but not precisely defined in the paper) that single-reward RLHF fails minority groups because of a fairness-utility trade-off (roughly: when a minority's reward is negatively correlated with the majority's, you cannot satisfy both). The authors argue against this by showing that even when the minority reward is positively correlated with the majority's, minority reward can still collapse. They then attribute the collapse to gradient conflict, contrasting a method without gradient surgery (MaxMin) against one with it (PCGrad), and showing the latter alleviates the collapse.

Summary Of Strengths:

The paper is clearly written and easy to follow, and its core experimental design is a genuine strength: the controlled α-sweep lets the authors vary preference divergence continuously and cleanly separate the regime where group rewards are still positively correlated from where they conflict. The use of PCGrad as a direct intervention (rather than relying on observational correlation alone) is a meaningful piece of causal evidence for the paper's central claim. The authors also engage honestly with known length-correlation effects in RLHF, both acknowledging the confound and taking concrete steps to mitigate it.

Summary Of Weaknesses:

W1 — Central "fairness-utility trade-off" claim is undefined and unattributed.

The paper is built around refuting the view that a fairness-utility trade-off is the reason for minority-group collapse. However, it never precisely states what the claim is, how it is specifically defined, or who holds it. As such, it is hard to assess whether they successfully refuted it.

Should not be our central claim. Double check where it is.

Collapse value level incompactable

W2 — Value-level vs. optimization-level distinction is unclear.

The paper's discussion revolves around distinguishing value-level from optimization-level conflict. The introduction attempts to define both, but the key difference between them and why it matters for the paper is unclear. Lines 56-58 ("...the preferences need not conflict; the RLHF updates can") seem to state a central argument, but there is not enough context or motivation to convey its significance; the "so what" is left implicit.

Explain what are they.

value -level -> scaling reward level

Optimization-level -> estimate advantage and change gradient

Decide how we refine RLHF: If value level, we need to reshape reward; if optimization-level, change how to optimize.

W3 — Causal claim not cleanly isolated.

The comparison provides some evidence, stronger than correlation, that gradient conflict drives the collapse. But this would only cleanly isolate the gradient mechanism if PCGrad and MaxMin differed

solely in whether gradient surgery is applied. They do not: they differ on at least two axes at once 1) whether gradients are projected, and 2) whether both objectives stay active, so PCGrad's advantage cannot be attributed to the projection mechanism specifically. A method that keeps both objectives active without gradient surgery, or one that applies surgery but serves one group at a time, would help disentangle these.

We choose to use the sum of two objective gradients as the optimization direction, which serves as a method of keeping both objectives active without gradient surgery. The second type of method does not fit this task. If there were one group and one objective, what surgery is required remains unclear.

W4 — Update-level interference: transient cause vs. persistent effect.

In Fig. 3, negative cosine similarity appears early but returns positive later, yet the performance collapse is persistent (Fig. 2). If the proposed cause disappears but the effect persists, the causal story needs to explain the gap. The paper acknowledges this ("the diagnostic is most informative about the onset of collapse") but does not explain why a transient conflict would produce a persistent collapse.

Early step only pcgrad, late step only pcgrad

W5 — Missing mix-ratio experiment.

This rests on the assumption that the aggregated reward is dominated by the majority. If so, the mix ratio should control collapse severity — yet the paper uses only one split (80/20). Sweeping it would directly test the mechanism: if collapse severity scales with majority proportion, that supports the majority-domination account.

Change proportion,

Comments Suggestions And Typos:
MaxMin-RLHF and PCGrad should be cited at first mention in the introduction, so readers aren't left unsure whether these are newly proposed methods. The subscript i in Equation 4 should be defined.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.
Soundness: 3.5
Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.
Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.
Limitations And Societal Impact:
Yes

Ethical Concerns:
None

Needs Ethics Review: No
Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Response 1

We thank the reviewer for the thoughtful assessment. We appreciate the recognition of the controlled α-sweep as it cleanly separates positively correlated from conflicting preference regimes, that PCGrad provides direct intervention-based evidence beyond correlation, and that we mitigate the length-correlation confound. We respond below.

W1 Definition and attribution of the "fairness-utility trade-off."

We will define the claim on first use. In this work, **utility** is population-weighted reward, and **fairness** is worst-group performance. A fairness–utility trade-off means that improving the worst-group reward necessarily reduces aggregate utility because the group objectives are intrinsically incompatible, from Thm 3.3 and Figure 3 in Chakraborty et al. (2024). Our results show that this explanation is insufficient in our setting: minority collapse occurs while the two group rewards remain positively correlated, so it cannot be explained solely by intrinsic value conflict. Instead, the pooled PPO update becomes anti-aligned with the minority objective. Thus, optimization-level interference is a key contributor to the failure, rather than evidence for an unavoidable fairness–utility trade-off.

Chakraborty, Souradip, et al. "MaxMin-RLHF: Alignment with Diverse Human Preferences." International Conference on Machine Learning. PMLR, 2024.

W2 Why the value-level vs. optimization-level distinction matters.

We will clarify both the distinction and its significance. **Value-level conflict** concerns reward geometry: improving one group's reward intrinsically harms another's. **Optimization-level interference** concerns update geometry: the group rewards remain positively correlated, yet the pooled PPO update moves against one group's gradient. This distinction matters because it determines the remedy. Value conflict motivates objective-level methods such as MaxMin-RLHF, whereas optimization interference motivates directly modifying conflicting update directions, as in PCGrad. We will make this "so what" explicit in the revised version.

W3 Isolating the gradient-surgery mechanism.

We implement the suggested control ****SumGrad****, which keeps both group objectives active and sums their gradients without projection. SumGrad is identical to PCGrad except for the projection step. PCGrad–SumGrad isolates the effect of gradient surgery, while SumGrad–MaxMin compares joint optimization of both group objectives against worst-group objective selection. Worst-group performance shows that keeping both objectives active is insufficient: for $\alpha \geq 1.2$, SumGrad still collapses, whereas PCGrad remains substantially more robust. Averaged over $\alpha \geq 1.0$, PCGrad improves worst-group reward over SumGrad by 0.39, while SumGrad improves over MaxMin by 0.03. We will include this control and revise our causal claim accordingly.

α	MaxMin	SumGrad	PCGrad
0.6	0.926	0.939	0.949

0.8	0.841	0.897	0.910
1.0	0.693	0.822	0.903
1.1	0.729	0.819	0.869
1.2	0.181	-0.106	0.835
1.3	-0.272	-0.073	0.432

W4 Transient Cause, Persistent Collapse

We explain why a transient conflict can cause persistent collapse using the following experiments. At $\alpha=0.75$, we optimize the same majority and minority reward, but enable PCGrad projection only during steps 1–400 or only during steps 401–800. Outside that window, the pooled PPO update is used. Early projection raises worst-group reward from 0.246 to 0.736, whereas late projection reaches only 0.307. This shows that early conflicting updates initiate the persistent collapse, while correcting them afterward provides little recovery. Because PPO is on-policy, early updates alter subsequent samples. Thus the later positive cosine reflects local alignment at an already-collapsed policy, not reversal of the earlier damage.

Method	Vanilla	PCGrad	Early	Late
Worst group performance	0.246	0.915	0.736	0.307

W5 Effect of Group-Mix Ratio

Following the reviewer’s suggestion, we sweep the minority proportion while fixing $\alpha=1.0$ and all other settings. At minority proportion 0.01–0.2, the majority reward remains near 0.99, while the minority reward stays close to zero. At 0.3 and 0.4, the minority reward improves and the gap between groups narrows. These results show that collapse severity is sensitive to the group-mix ratio and are consistent with majority domination.

Minority prop.	minority	majority
0.01	0.065	0.990
0.1	0.074	0.990
0.2	0.057	0.992
0.3	0.125	0.712
0.4	0.340	0.707

Minor Comments and Typos

We thank the reviewer for these helpful suggestions. We will cite MaxMin-RLHF and PCGrad at their first mention and explicitly define the index i in Equation 4 as the group index.

jxS1-2.5

Official Review of Submission10802 by Reviewer jxS1

Official Reviewby Reviewer jxS103 Jul 2026, 02:33 (modified: 08 Jul 2026, 15:46)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer jxS1, AuthorsRevisions

Paper Summary:

This paper studies heterogeneous preference RLHF in a controlled two group setting. It shows that pooled RLHF can suffer a sharp worst group collapse even when the group rewards remain positively correlated, and claims this is due to gradient interference rather than an inherent fairness/utility trade-off. It compares vanilla RLHF, MaxMin RLHF, and PCGrad-PPO, concluding that PCGrad recovers worst group performance better by directly modifying conflicting group gradients.

Summary Of Strengths:

1. The paper studies an interesting question: whether worst group collapse in single reward RLHF can come from gradient interference, rather than only value conflict between group preferences.
2. The paper is clear that this is a controlled two-group study, not a general solution for all heterogeneous preferences. The controlled setting changes the preference divergence between the two groups, and shows that the collapse can happen while the two rewards are still positively correlated.
3. The connection to gradient-based MOO is interesting. The comparison between MaxMin-RLHF and PCGrad-PPO suggests that modifying update directions may be better than changing the scalar objective.

Summary Of Weaknesses:

1. The main method seems to be just applying PCGrad to PPO with two group rewards. I do not see a deeper analysis showing if RLHF has a special structure that prefers some gradient-based MOO methods over others. The empirical finding is interesting, but currently the contribution feels a bit narrow. Also, there are many exisiting methods in gradient-based MOO field that can yield non-conflicting directions, i.e. resolving gradient-conflict (eg, Nash-MTL, MGDA, DualProjection), why this paper only considers PCGrad?
   We are not proposing new methods. Diagnostic paper.

   On-policy, error accumulates. Tricks in PPO (clipping)

2. Only two groups is a special case in multi-objective optimization (MOO). For two objectives, PCGrad reduces to a simple two-objective projection rule. The more difficult issues appear for three or more objectives: PCGrad can depend on projection order/randomness, and it does not in general guarantee a direction that is non-conflicting for all objectives. Since heterogeneous preferences can naturally involve more than two groups, it is unclear how much the conclusion extends beyond this special case. The paper should discuss this limitation more, a 3 (or more) groups study would make the contribution more convincing.

   Not do experiments.

Comments Suggestions And Typos:

Is the effect of PCGrad fully separated? Vanilla RLHF uses one reward model trained on aggregated preference data, while PCGrad-PPO keeps separate reward models and computes separate group gradients before applying projection. Thus, the comparison changes both the reward-model setup and the update rule. A useful baseline could be a group-aware PPO method using the same separate reward models and the same setup as PCGrad-PPO, but simply combining the two group

gradients without projection? This would clarify whether the gain comes from the PCGrad projection itself, or more generally from keeping the group objectives separate.

Evaluate the accuracy of reward model.

The method proposed does not make sense to me. But we can try.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.
Soundness: 2.5
Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.
Overall Assessment: 2.5 = Borderline Findings
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Response 2

We thank the reviewer for the careful reading and constructive feedback. We agree that the original presentation should better distinguish our diagnostic contribution from the choice of PCGrad as one intervention. We address the comments below.

W1 — Broader Gradient-Based MOO Baselines and Scope

We clarify that this is primarily a diagnostic study, rather than a proposal of PCGrad-PPO as a universally superior RLHF method. The RLHF-specific object we study is the stochastic, on-policy, KL-regularized PPO update induced by learned group rewards. Our central claim is that this pooled update can harm one group even before the group rewards become negatively correlated. PCGrad was chosen as a simple intervention that directly removes conflicting gradient components.

We agree that evaluating only PCGrad makes the empirical scope appear narrow. Following the reviewer's suggestion, we add MGDA-PPO as an additional gradient-based MOO baseline. Its worst-group performance is comparable to PCGrad-PPO across most α values and better at $\alpha=1.3$. This shows that the improvement is not specific to PCGrad and supports the broader conclusion that explicitly correcting update geometry can mitigate the collapse. We will position PCGrad as one representative gradient-level correction rather than as the uniquely preferred method for RLHF.

α	MaxMin	SumGrad	PCGrad	MGDA
0.6	0.926	0.939	0.949	0.937
0.8	0.841	0.897	0.910	0.924
1.0	0.693	0.822	0.903	0.912
1.1	0.729	0.819	0.869	0.825
1.2	0.181	-0.106	0.835	0.842
1.3	-0.272	-0.073	0.432	0.536

W2 — Extension Beyond Two Groups

We use two groups to isolate the failure mechanism: a pooled PPO update can harm one group even when the group rewards remain positively correlated, revealing optimization-level interference rather than intrinsic reward conflict.

We do not claim that two-group PCGrad extends unchanged to three or more groups. With multiple objectives, PCGrad can depend on projection order and does not guarantee a direction that benefits every group. Thus, the broader implication is not that PCGrad is always the appropriate solution, but that heterogeneous-preference RLHF should explicitly examine and correct update geometry. We will add this limitation and identify multi-group evaluation with methods designed for several objectives as important future work.

Comment — Isolating the gradient-surgery mechanism.

We thank the reviewer for this suggestion. We add the suggested **SumGrad**, which uses the same separate reward models, group-specific PPO objectives, KL regularization, and value loss as PCGrad-PPO, but directly sums the two group gradients without projection.Thus, PCGrad–SumGrad isolates the contribution of the projection step from that of maintaining separate group objectives. The results in W1 show that separation alone is insufficient: for $\alpha \geq 1.2$, SumGrad still exhibits severe worst-group collapse, whereas PCGrad remains substantially more robust. Averaged over $\alpha \geq 1.0$, PCGrad improves over SumGrad by 0.39. We will include this baseline and refine our causal claim accordingly.

8Gy6-3

Official Review of Submission10802 by Reviewer 8Gy6

Official Reviewby Reviewer 8Gy601 Jul 2026, 08:15 (modified: 08 Jul 2026, 15:46)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 8Gy6, AuthorsRevisions

Paper Summary:

This paper investigates group-aware RLHF under heterogeneous preferences. Its core argument is that poor alignment performance for minority groups does not always stem from unavoidable value-level conflicts. Instead, it can also arise from optimization-level interference: policy updates induced by the aggregated objective harm certain groups even when the rewards of different groups remain broadly compatible overall. To verify this claim, the paper constructs a controlled RLHF environment based on the IMDb dataset with two synthetic preference groups. Each prompt-response pair is scored along two dimensions: positive sentiment and conciseness. The majority group prioritizes sentiment, whereas the minority group values conciseness more. A hyperparameter governs the degree of preference divergence between the two groups, ranging from perfectly identical preferences to increasingly conflicting ones. The authors compare vanilla single-reward RLHF, MaxMin-RLHF, and PCGrad-PPO. Key experimental results show that vanilla RLHF can suffer severe minority/worst-group collapse even when group rewards are positively correlated. Meanwhile, PCGrad-PPO delivers stronger and more consistent improvements to worst-group performance relative to MaxMin-RLHF. This indicates that some group alignment failures should be framed as gradient interference failures rather than inherent value conflicts.

Summary Of Strengths:

1. The paper makes a useful distinction between value-level conflict and optimization-level interference. This is a helpful framing for group-aware RLHF, where poor minority-group performance can be too quickly attributed to irreconcilable preference conflict.
2. The controlled IMDb setup is simple and interpretable. By varying , the paper creates a continuous path from aligned preferences to increasingly divergent preferences, which makes the collapse regime easier to analyze.
3. The gradient-alignment diagnostic is a strong part of the paper. Measuring the cosine similarity between the PPO update and the minority-group gradient gives more direct evidence for the optimization-interference interpretation than final group rewards alone.
4. The comparison between MaxMin-RLHF and PCGrad-PPO is well motivated because the two methods intervene at different levels: MaxMin changes the scalar objective, while PCGrad modifies the update geometry.
5. The appendix provides useful implementation details on the IMDb setup, reward attributes, KL handling, PPO configuration, MaxMin implementation, and PCGrad-PPO gradient composition.

Summary Of Weaknesses:

1. The experimental setup is highly simplified. The primary empirical evidence is derived from an IMDb review generation task with two manually designed reward attributes: sentiment and conciseness. While this design serves controlled diagnostic purposes well, real-world preference heterogeneity typically involves far more dimensions, noisy human judgments, and context-dependent preferences.

   Example experiment, based on which maxmin-rlhf is tested.

2. The experiments adopt GPT-2 instead of modern instruction-tuned large language models. Given the paper's motivation centered on group-aware RLHF for LLMs, it remains unclear

whether identical gradient interference patterns and performance gains from PCGrad-PPO generalize to larger instruction-tuned models, contemporary RLHF/RLAIF pipelines, or other mainstream optimization frameworks.

LLAMA

3. The suite of compared baselines is limited. The paper only benchmarks vanilla RLHF, MaxMin-RLHF, and PCGrad-PPO, omitting other widely adopted multi-objective and group-aware optimization techniques such as GradNorm, CAGrad, MGDA/Pareto multi-task learning, group-weighted PPO, and tuned scalarization approaches.

   Try implementing GradNorm

   We are not presenting sota method. We are presenting root cause.

4. The two-group experimental design restricts the generalizability of the conclusions. Real-world preference diversity often consists of numerous distinct groups or continuous distributions of user preferences. As the number of groups expands, PCGrad introduces additional practical complexities. The manuscript ought to discuss computational overhead, sensitivity to gradient projection order, imbalanced gradient magnitudes, and group sampling challenges under multi-group configurations.

   Only discuss.

5. Qualitative analysis of model generations is absent. Since all reward signals are synthetic, presenting sample outputs across varying values and across vanilla RLHF, MaxMin-RLHF, and PCGrad-PPO would help validate whether PCGrad-PPO genuinely produces higher-quality responses, rather than merely overfitting to the artificially constructed reward functions

   Find some

Comments Suggestions And Typos:

1. Further explain why positive Pearson correlation suffices as evidence of compatible preferences. Pareto front analysis or joint improvement tests would strengthen the argument regarding value-level conflicts.

   Ask gpt: what others can we use as evidence.

2. Add experiments with at least one larger or modern instruction-tuned model. Even a smaller Llama or Pythia variant would greatly improve external validity compared to GPT-2.
3. Provide generated response examples for Vanilla, MaxMin, and PCGrad at representative values, especially around = 0.7, 0.75, 1.0, and 1.2.
4. Discuss how PCGrad-PPO scales to more than two groups, covering computational overhead, sensitivity to gradient projection order, imbalanced gradient magnitudes, and group sampling strategies.
5. Minor writing issues: Standardize the naming of the method uniformly across the paper: use either "MaxMin-RLHF" or "Max-Min RLHF" consistently. Provide an explicit definition of "worst-group collapse" before its first occurrence in the results section.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.
Excitement: 3.5
Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.
Ethical Concerns:
There are no concerns with this submission

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.
Datasets: 2 = Documentary: The new datasets will be useful to study or replicate the reported research, although for other purposes they may have limited interest or limited usability. (Still a positive rating)
Software: 2 = Documentary: The new software will be useful to study or replicate the reported research, although for other purposes it may have limited interest or limited usability. (Still a positive rating)
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Response 3

We thank the reviewer for the constructive feedback. We appreciate their recognition of the controlled α-sweep and gradient-alignment diagnostic. We address the comments below.

W1 — Simplified Experimental Setting

The IMDb setup follows the benchmark introduced with MaxMin-RLHF and is designed to isolate why vanilla RLHF fails as preference divergence increases. We agree that real-world preferences are higher-dimensional, noisy, and context-dependent, and therefore do not claim that this benchmark captures their full complexity. Our broader conclusion is methodological: when failure arises from gradient interference, rescaling reward is insufficient, and updating geometry on optimization-level should be addressed. We will make this scope more explicit.

W2 — Generalization Across Model Backbones

We will add experiments with **Pythia-160M** and **Llama3.2-1B** as both reward-model and policy backbones. At $\alpha=1.0$, vanilla RLHF again yields near-zero minority reward, while MaxMin-RLHF and PCGrad-PPO substantially recover it on both models. Although absolute majority scores vary, the minority-group improvement is consistent across backbones, indicating that the observed failure and benefit of gradient-aware optimization are not specific to GPT-2.

| Model | Method |Minority |Majority |

|-----|--------|---------|--------|
Pythia	Vanilla	0.076	0.653
Pythia	MaxMin	0.957	0.619
Pythia	PCGrad	0.968	0.654
LLAMA	Vanilla	0.027	0.684
LLAMA	MaxMin	0.902	0.695
LLAMA	PCGrad	0.956	0.705

W3 — Broader MOO Baselines

This paper is primarily diagnostic rather than a proposal of PCGrad as the uniquely best optimizer. Following the reviewer's suggestion, we add **MGDA-PPO** as a complementary gradient-based MOO method. Its worst-group performance is comparable to PCGrad-PPO across most α values and better at $\alpha=1.3$, supporting our broader claim that modifying update geometry can outperform reward-level reweighting alone.

α	MaxMin	PCGrad	MGDA
0.6	0.926	0.949	0.937
0.8	0.841	0.910	0.924
1.0	0.693	0.903	0.912
1.1	0.729	0.869	0.825
1.2	0.181	0.835	0.842

| 1.3 | -0.272 | 0.432 | 0.536|

W4 — Practical Challenges in Multi-Group Settings

We use two groups to isolate the gradient-interference mechanism, rather than to claim that PCGrad is the definitive solution for arbitrary numbers of groups. With K groups, computing group-specific gradients increases optimization and memory costs, while pairwise projection scales as $O(K^2)$. PCGrad can also depend on projection order and does not guarantee a direction that benefits every objective when $K>2$. Projection alone also does not resolve imbalanced gradient magnitudes or unequal group sampling. Multi-group systems may therefore require order-invariant MOO methods, gradient normalization, and balanced or importance-weighted sampling. We will add these limitations. Our broader finding is that update conflicts should also be diagnosed and corrected in multi-group settings, although the appropriate optimizer may differ from PCGrad.

W5 — Qualitative Generations

We will add representative outputs for vanilla RLHF, MaxMin-RLHF, and PCGrad-PPO at $alpha\in\{0.7,0.75,1.0,1.2\}$, together with their sentiment, conciseness, and length statistics. For example, at $\alpha=0.75$:

- Vanilla: “This movie is not that interesting, but we are delighted to be bringing this exciting ensemble…”
- MaxMin: “This movie is not that interesting, and this is just a collection of stunning artwork.”
- PCGrad: “This movie is not the best, but it is a great experience!”

These examples suggest that PCGrad better balances the two designed attributes.

W6 — Evidence Beyond Pearson Correlation

We agree that positive Pearson correlation alone does not establish Pareto compatibility. We therefore add a local joint-improvement test using MGDA at the vanilla checkpoint near collapse. The MGDA direction is positively aligned with both group gradients, and a small update increases both majority and minority rewards. The resulting policy Pareto-dominates the collapsed vanilla policy in the group-reward plane. We will retain Pearson correlation as a global descriptive statistic, while using this joint-improvement result as more direct evidence that the collapse is not required by value-level conflict.

Minor Writing Issues

We thank the reviewer for identifying these presentation issues. We will standardize the method name as **MaxMin-RLHF** throughout the paper and define **worst-group collapse** before its first use as a sharp deterioration in the minimum performance across groups while other group performance remains comparatively high.

Summary to AC

Dear AC and SAC,

We sincerely appreciate your time and effort in overseeing the review process for our submission. We understand that ACs are facing significant additional workloads this year, and we would like to share a brief summary of our rebuttal to assist in your decision-making.

> Aligning a single policy with heterogeneous preferences is a challenge in RLHF. Poor worst-group performance in single-reward RLHF is often attributed to value-level incompatibility, under which improving one group's reward necessarily harms another's. Our paper identifies a distinct and **previously underexplored optimization failure**: worst-group collapse can arise from gradient interference **even while group rewards remain positively correlated**. In this regime, directly addressing conflicts among group gradients is more effective than changing the scalar objective alone. By separating these two failure modes, our analysis provides a clearer basis for future group-aware alignment methods to **match the remedy to the source of failure**.

1. **We appreciate the reviewers' recognition of our work's strengths.**

Across the three reviews, several strengths were acknowledged:

(i) Controlled and interpretable experimental design with strong empirical evidence.

8R12: "The controlled α-sweep lets the authors vary preference divergence continuously and cleanly separate the regime where group rewards are still positively correlated from where they conflict."
jxS1: "The controlled setting …. shows that the collapse can happen while the two rewards are still positively correlated."
8Gy6: "The **gradient-alignment diagnostic** is a strong part of the paper."

(ii) Well-motivated gradient-based extension with clear improvements
8R12: "The use of PCGrad as a direct intervention … is a meaningful piece of causal evidence for the paper's central claim."
8Gy6: "The comparison between MaxMin-RLHF and PCGrad-PPO is **well motivated** because the two methods intervene at different levels: scalar objective and update geometry."

(iii) Clear and easy-to-follow presentation
8R12: "The paper is clearly written and easy to follow."
jxS1: "The paper is clear that this is a controlled two-group study, not a general solution for all heterogeneous preferences."

2. **We clarify that this is a **diagnostic paper rather than a method proposal**.**

We use PCGrad as one intervention to demonstrate that correcting update geometry can mitigate gradient interference, but we do not claim that PCGrad is universally optimal or extends unchanged to many groups. Our broader conclusion is that multi-group RLHF should explicitly diagnose optimization-level conflicts and adjust the update direction accordingly.

3. **We conducted additional rebuttal experiments and clarifications to address the remaining concerns.**

Guided by the reviewers' suggestions, we conducted additional experiments to directly address the main concerns about baseline coverage, model scale, and the causal claim:

(i) MGDA-PPO: We added another gradient-based MOO method, showing that the improvement of correcting update geometry is not specific to PCGrad, supporting the generality of our conclusion.

(ii) SumGrad: We added a control that isolates the effect of gradient surgery from that of maintaining separate objectives. The results confirm that the reported improvement comes primarily from gradient surgery rather than from keeping both objectives active.

(iii) Additional model backbones: We extended the experiments from GPT-2 to Pythia-160M and Llama3.2-1B as both policy and reward models. The consistent findings show that the diagnosed failure and improvement of gradient-aware optimization are not specific to GPT-2.

We sincerely thank all reviewers for their thoughtful comments and suggestions. We prepared detailed responses to the main concerns raised in the reviews and added corresponding experiments and clarifications. Although the opportunity for further discussion may be limited at this stage, we are confident that our comprehensive rebuttal directly addresses the key concerns raised in the reviews.

We hope this summary is helpful in facilitating your decision. All points in this summary are based on objective information from the rebuttal. Once again, we thank you and all the reviewers for your thoughtful evaluation and for helping us improve this work.

With best wishes,
The Authors

~~(iv) Mechanism-level controls: We ran two experiments testing the causal claim directly. A minority-proportion sweep shows collapse severity scales with majority share; early-vs-late gradient projection explains why a transient conflict causes persistent collapse.~~

4. **We also addressed conceptual concerns about the framing. **

Reviewer 8R12 noted the paper refutes a prior "fairness-utility trade-off" view without defining it. Our contribution does not depend on that refutation: we directly show the collapse comes from gradient interference and that fixing the gradients recovers the worst group better than rescaling the reward does, shown by intervention rather than by characterizing prior work. The revision defines the terms precisely and clarifies why the distinction matters for our result.

Response to Meta

We thank the meta-reviewer for the careful assessment. We clarify that our paper does not claim that worst-group performance is the uniquely correct fairness objective, or that PCGrad selects a universally “better” Pareto optimum. Our question is narrower: is vanilla RLHF’s collapse an unavoidable reward trade-off, or an avoidable optimization failure?

In our setting, vanilla RLHF uses pooled preference data and optimizes a single reward, and group-aware methods additionally use group labels and separate rewards. We define utility as population-weighted reward and fairness as worst-group reward, and report minority, majority, and aggregate performance so that the cost of improving the worst group is explicit.

Crucially, our result is not merely that different algorithms reach different Pareto optima. Following the reviewers’ suggestion, we initialize MGDA from the collapsed vanilla checkpoint. MGDA finds a direction positively aligned with both group gradients, and a small update improves both rewards. Thus, the collapsed vanilla policy is locally Pareto-dominated rather than simply another Pareto optimum.

PCGrad is a diagnostic intervention, not a universal solution. We added MGDA-PPO to support the broader conclusion that correcting update geometry matters, not to establish that either method is universally superior.

We will revise the framing accordingly: this is a controlled diagnostic study of one identifiable failure mode of pooled RLHF, not a claim that all heterogeneous-preference problems reduce to gradient conflict.